

OPINION

New roles for cyclin-dependent kinases in T cell biology: linking cell division and differentiation

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Abstract | The proliferation of a few antigen-reactive lymphocytes into a large population of effector cells is a fundamental property of adaptive immunity. The cell division that fuels this process is driven by signals from antigen, co-stimulatory molecules and growth factor receptors, and is controlled by the cyclin-dependent kinase (CDK) cascade. In this Opinion article, we discuss how the CDK cascade provides one potential link between cell division and differentiation through the phosphorylation of immunologically relevant transcription factors, and how components of this pathway might ultimately participate in the decision between tolerance and immunity.

Upon recognition of their specific antigen, lymphocytes are capable of massive clonal expansion. For example, early advances in the detection of effector cytokine production at the single-cell level showed that lymphocytic choriomeningitis virus (LCMV)-specific cytotoxic T lymphocytes (CTLs) constitute up to 70% of the total CD8⁺ T cell population in mice at the peak of the antiviral response^{1,2}. This equates to at least 15 cell divisions per initial precursor cell, to a doubling time of approximately 14 hours and to a population expansion of nearly 30,000-fold. More recent studies using MHC-peptide tetramers to identify antigen-specific CD4⁺ and CD8⁺ T cells within a polyclonal repertoire agree with these metrics, showing that a population of ~200 naive T cells can expand ~50-fold upon antigenic stimulation in only 4 days, with doubling times ranging from 12 hours to 16 hours, depending on the antigen³. In the mid-1990s, immunologists began to use fluorescent dyes that segregate uniformly between daughter cells to track individual B cell and T cell mitoses during clonal expansion *in vitro* and *in vivo*^{4–7}. This powerful approach showed that antigen-specific CD4⁺ T cells undergo the first mitosis

~30 hours after activation *in vivo*, and have an average doubling time of 10 hours during the peak phase of clonal expansion, with some cells in the population transiting the cell cycle in less than 6 hours^{5,7}. These rates of clonal expansion put a tremendous demand on cellular anabolic capacity and on the fidelity of the cell cycle machinery.

The cell division cycle is coordinated by a series of cyclin-dependent kinases (CDKs) and their inhibitory partners (BOX 1). This Opinion article highlights the impact of this regulatory network on T cell anergy, differentiation, memory and tolerance. We propose a model to explain the link between cell division and effector T cell differentiation that focuses on newly discovered, cell cycle-independent roles for CDKs as regulators of immunologically relevant transcription factors. Finally, we discuss the potential of this model in providing novel pharmacological targets for immunosuppressive therapy in the clinic.

The CDK cascade

The CDKs are a family of proline-directed serine/threonine kinases that were first identified as regulators of cell cycle progression (see REF. 8 for a recent review). Over the

past decade this family has grown to include 20 members: CDK1 is the ancestral mitotic kinase, and CDK2, CDK4 and CDK6 regulate progression through interphase (BOX 1). CDK5 does not seem to regulate the cell cycle, but instead controls the development and function of neurons. CDK7, CDK8 and CDK9 regulate RNA polymerase II (RNA Pol II)-dependent transcriptional initiation and elongation, and CDK11 is involved in mRNA splicing. The functions of CDK3, CDK10, and CDK12 to CDK20 are not well known, but these kinases seem to have general non-cell cycle roles in DNA damage repair, transcription and signal transduction. The activity of CDKs is controlled by the availability of their cyclin partners, and by physical interactions with members of the inhibitor of CDK4 (INK4) family, such as p18 (also known as INK4C; which is encoded by *CDKN2C*); and of the kinase inhibitory protein (KIP) family, such as p27 (also known as KIP1; which is encoded by *CDKN1B*).

Tissue-specific roles for interphase CDKs in development. The classical view of cell cycle regulation (BOX 1), in which each interphase CDK is absolutely required for progression past a distinct cell cycle checkpoint, was mostly derived from *in vitro* studies of kinase-dead CDK mutants and CDK inhibitors overexpressed in fibroblasts and transformed epithelial cell lines^{9,10}. However, more recent approaches in which individual CDKs were genetically deleted in mice have led to a reassessment of this scenario. Deletion of *Cdk1* in mice results in lethality before embryonic day 2.5 (E2.5), which confirms that this kinase is absolutely required for mitosis in all tissues^{11,12}. However, mice with a deletion of *Cdk2* (REFS 13, 14), *Cdk4* (REFS 14–16) or *Cdk6* (REF. 17) are viable and have generally normal development, with some effects on specific tissues. For example, CDK2 is required for meiosis, and mice lacking *Cdk2* are sterile owing to defects in sperm and ovary development¹⁴, whereas CDK4 is essential for the normal proliferation of pancreatic β -cells and pituitary lactotrophs¹⁶. Mice deficient in CDK6 or its catalytic partner cyclin D3 are viable

and have normal development, except for a marked decrease in thymocyte numbers owing to the failure of Notch-dependent proliferation at the DN3 to DN4 transition^{17–19}. Thymocyte proliferation requires CDK6 activity, as mice transgenic for a kinase-dead mutant of CDK6 have a similar phenotype to CDK6-deficient mice²⁰. This CDK6–cyclin D3-driven proliferation of thymocytes is controlled by the D-type CDK inhibitor p16 (also known as INK4A; which is encoded by *CDKN2A*), as a lack of p16, but not of other members of the INK4 family²¹, results in hyperplasia of the double-positive (CD4⁺CD8⁺) thymocyte compartment²² (FIG. 1). Defects in CDK6 activity also render thymocyte precursors resistant to AKT- and LCK-driven leukemic transformation^{18,19}, which shows that CDK6–cyclin D3 operates downstream of LCK, Notch and AKT to drive both normal and malignant thymocyte proliferation.

Together, these genetic studies indicate that the interphase CDKs have a redundant role in some tissues, but not in others. Such redundancy has been shown between CDK2 and CDK4 in the heart. Deletion of either CDK alone does not affect cardiac development, but double deficiency results in defective cardiomyocyte development and embryonic lethality²³, which can be rescued if these CDKs are deleted after cardiac development²⁴. Remarkably, mice lacking CDK2, CDK4 and CDK6 survive until E15.5 (REF. 12), at which time they die owing to defects in the formation of hepatic, cardiac and haematopoietic

Box 1 | The classical view of CDKs in cell cycle progression

During the cell cycle (FIG. 2), a series of sequential cyclin-dependent kinase (CDK)–cyclin-directed phosphorylation events enable the cell to pass crucial cell cycle checkpoints, resulting in cell growth, DNA replication and mitosis^{21,92}. Mitogenic stimuli drive cells from G0 phase into G1 phase by inducing the expression of D-type cyclins, which activate CDK4 and CDK6 by competing for their binding to the CDK inhibitors p16 (also known as INK4A; which is encoded by *CDKN2A*), p15 (also known as INK4B; which is encoded by *CDKN2B*) and p18 (also known as INK4C; which is encoded by *CDKN2C*). CDK4 and CDK6 phosphorylate and inactivate retinoblastoma protein (RB), a negative regulator of E2F transcription factors, leading to the induction of proteins required for DNA replication, and to increased levels of cyclin E, the partner of the major G1/S phase kinase CDK2. The mitogen-activated protein kinases (MAPKs), SRC family kinases, Janus kinases and phosphoinositide 3-kinases (PI3Ks) cooperate with CDK2–cyclin E to target the G1 phase inhibitor p27 (also known as KIP1; which is encoded by *CDKN1B*) for degradation, leading to the phosphorylation and activation of factors that promote histone synthesis, centrosome duplication and the assembly of DNA polymerase complexes at origins of DNA replication. During S phase, CDK2 partners with cyclin A to inhibit further E2F activity and replication origin assembly, which ensures that DNA replication occurs only once before mitosis. CDK1 then partners with cyclin B, which is induced during S phase, to promote nuclear envelope breakdown, chromatin de-condensation, mitotic spindle assembly and cell division. Experimental inhibition of individual CDK–cyclin activities results in arrest at distinct cell cycle checkpoints. Inappropriate CDK activity caused by cyclin overexpression or loss of CDK inhibitor function can bypass these checkpoints, leading to dysregulated DNA replication and mitosis, culminating in either cell death or cancer.

tissues. Embryonic fibroblasts derived from these mice cycle normally — albeit more slowly than wild-type cells — and they can be immortalized through standard oncogenic mechanisms. These data emphasize that CDK1 is the only CDK that is absolutely required to execute the general mitotic cell cycle programme, and they indicate that, although the interphase CDKs contribute to cell cycle progression, their only prerequisite role in development is in driving the differentiation of specific tissues.

Coupling differentiation to the cell cycle

A remarkable finding from early single-cell division tracing studies was that the acquisition of markers of lymphocyte differentiation and the capacity to secrete effector cytokines correlated with the number of cell divisions achieved during primary stimulation^{25–28}. T cells that complete more rounds of cell division during the primary response are more likely to produce effector cytokines upon secondary challenge. Conversely, T cell receptor (TCR)- and CD28-activated precursor T cells that achieve few or no cell divisions do not secrete cytokines, such as interferon- γ (IFN γ), that are typical of differentiated cells. Notably, such cells also fail to produce interleukin-2 (IL-2), a cytokine that even naive T cells can produce²⁹. This loss of function is a hallmark of T cell anergy³⁰, which indicates that a failure to divide following activation does not merely result in a lack of differentiation, but that it also induces tolerance. Anergy can be induced in TCR and CD28 co-stimulated CD4⁺ T cells by neutralization of IL-2 or by pharmacological inhibitors that arrest cells in late G1 phase^{31–33}. Inhibitory drugs that arrest the cell cycle at a later stage and allow cells to progress from G1 into S phase or G2 phase do not induce anergy. Thus, in addition to signalling through the TCR and CD28, T cells require further signals to pass an anergic checkpoint that is located downstream of the IL-2 receptor and that is also coupled to cell cycle control (reviewed in REF. 34).

Glossary

Anergy

The functional non-responsiveness of antigen-specific T cells.

Autoimmune accelerator locus

(*Yaa*). A duplication of a 4 Mbp segment of the Y chromosome of inbred BXSB mice containing 19 genes, including *Tlr7*, that leads to the spontaneous development of systemic autoimmunity similar to systemic lupus erythematosus.

Bioisosteres

In drug design, compounds that have highly similar properties and structures, but varied efficacy, activity or toxicity as a result of minor chemical substitutions.

DN3 to DN4 transition

The transition during thymocyte maturation during which CD4⁺CD8⁺ double-negative cells undergo T cell receptor- β (TCR β) rearrangement and downregulate CD25 expression.

G0 phase

(Resting phase). The period of the cell cycle during which the cell is quiescent that lasts until the initiation of cell division.

G1 phase

(Gap 1 phase). The phase of the cell cycle during which the cell grows and carries out large amounts of protein synthesis.

Interphase

The period representing the majority of the cell cycle including G1, S and G2 phases, during which the cell increases in size and replicates its DNA.

Interphase CDK

Cyclin-dependent kinases that are active during the interphase (G1, S and G2 phases) of the cell cycle, including CDK2, CDK4 and CDK6.

Nuclear mediator complex

The complex of multiple nuclear proteins that is required to enhance RNA polymerase II-mediated gene transcription.

RNA polymerase II

(RNA Pol II). A 550 kDa, 12-subunit eukaryotic enzyme that is required for the initiation of DNA transcription.

S phase

(Synthesis phase). The phase of the cell cycle during which the genomic DNA is replicated.

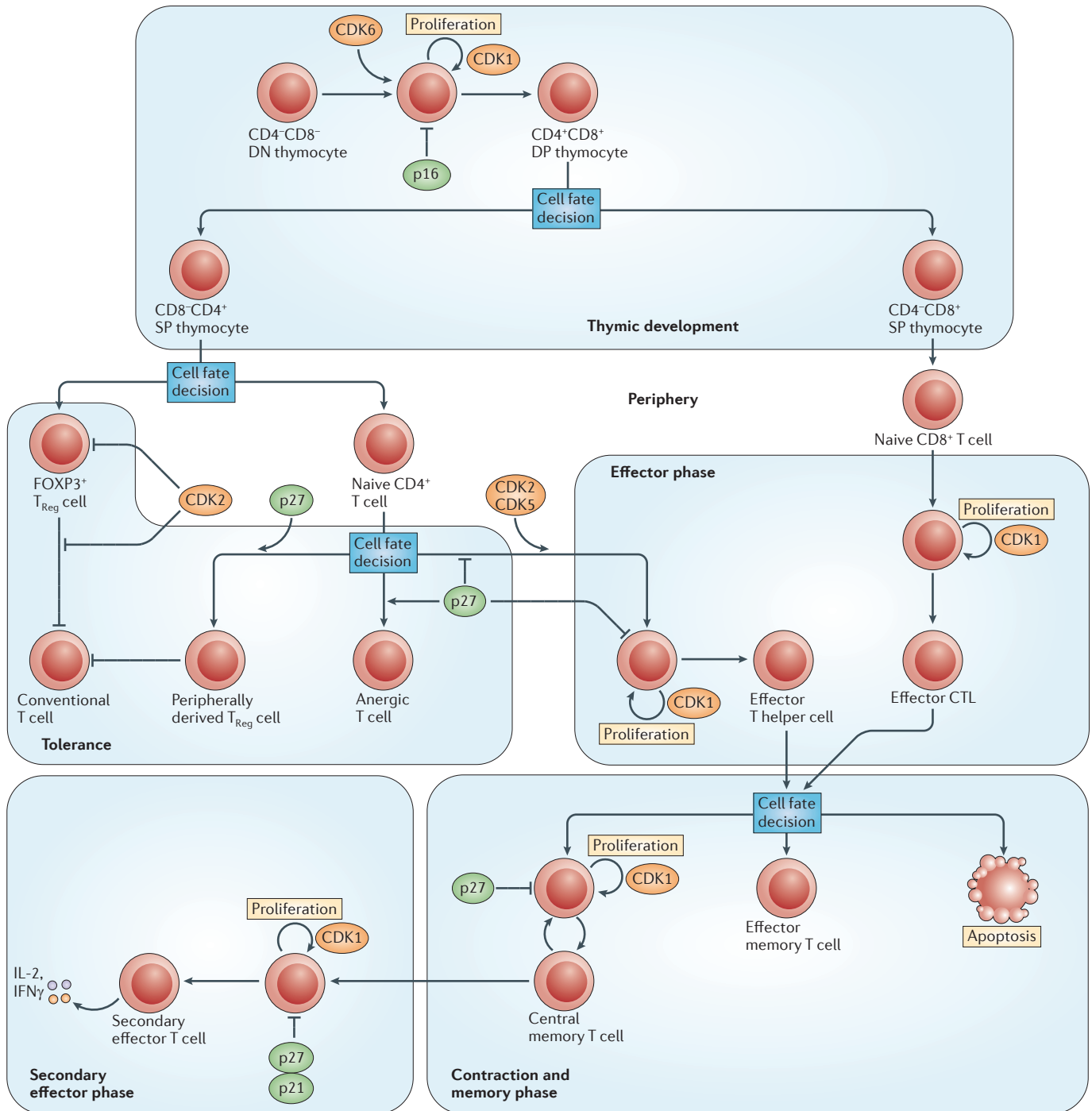


Figure 1 | Regulation of T cell development and differentiation by members of the CDK family. The figure shows the role of cyclin-dependent kinases (CDKs) and CDK inhibitors in the positive and negative regulation of T cell proliferation and cell fate decisions during thymic development, tolerance

induction, effector differentiation, contraction and memory formation, and restimulation (secondary effector phase). DN, double negative; CTL, cytotoxic T lymphocyte; DP, double positive; FOXP3, Forkhead box P3; IFN γ , interferon- γ ; IL-2, interleukin-2; SP, single positive; T_{Reg} cell, regulatory T cell.

Three non-mutually exclusive models for the link between T cell division and differentiation have developed over the past decade, the relative importance of which is not clear at present. One model proposes that S phase and mitosis function as a 'window of opportunity' for the epigenetic

modification of T cell effector genes. A second model suggests that signalling molecules and lineage-specific transcription factors segregate asymmetrically during cell division. A third model postulates that the factors involved in effector and/or memory cell gene expression programmes

are directly regulated by CDKs and/or other cell cycle-coupled kinase pathways (FIG. 2). Evidence exists for all three of these models. Owing to space constraints we do not cover the first two models in detail here, but they are summarized in BOX 2. In support of the first model, T cell differentiation

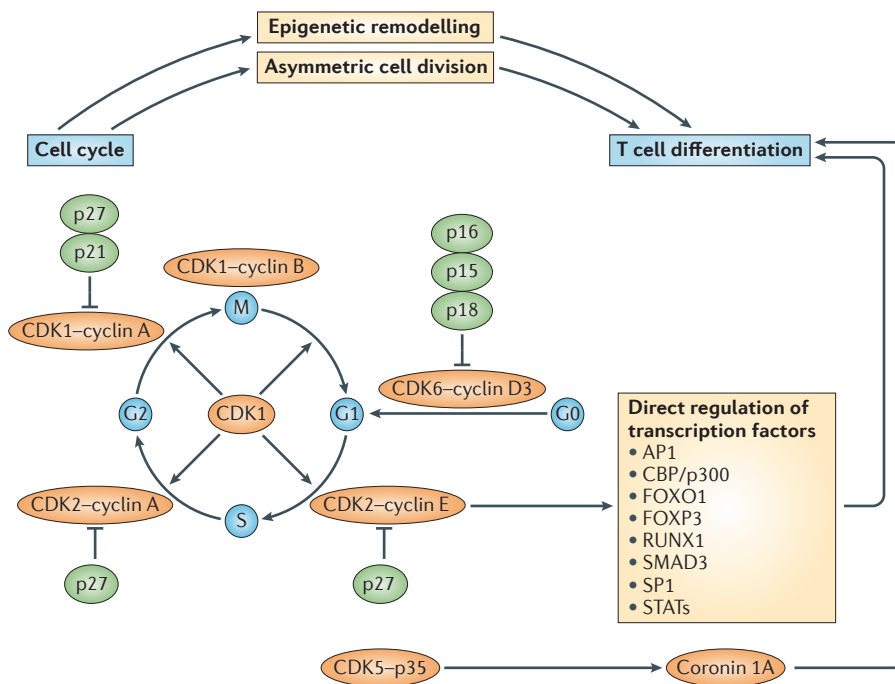


Figure 2 | Models to link cell cycle progression with T cell differentiation. Differentiation might be influenced by cell cycle-dependent epigenetic changes, the unequal (asymmetric) segregation of effector molecules between daughter cells during mitosis or, as we propose in this Opinion article, the direct regulation of transcription factors by cyclin-dependent kinases (CDKs). CBP/p300, CREB-binding protein; FOXO1, Forkhead box P3; SP1, transcription factor SP1; STATs, signal transducers and activators of transcription.

and anergy are regulated by DNA methylation and chromatin remodelling^{35,36}, and small-molecule inhibitors that promote an open chromatin conformation can uncouple the production of cytokines that are typical of differentiated T cells from cell division in activated T cells²⁸. In support of the second model, components of the TCR signalling synapse and transcription factors such as T-bet have been shown to segregate unevenly between daughter cells, thereby driving polarized cell fates within populations of proliferating T cells^{37–39}. By promoting cell division, the CDK cascade can drive these two processes. In this Opinion article, we focus on the experimental evidence for the third model, in which it is postulated that CDKs directly regulate factors involved in T cell differentiation.

p27 is an anergy versus differentiation

checkpoint sensor. Quiescent T cells express high levels of p27, which binds to and inhibits the activity of the interphase CDKs that are responsible for cell cycle progression. Upon TCR and CD28 co-stimulation, SRC kinases and the CD28-dependent kinase AKT phosphorylate p27, which decreases its affinity for CDK–cyclin complexes and

causes it to move from the nucleus to the cytoplasm^{40–42}. CD28 co-stimulation also induces expression of cyclin E, which activates CDK2 to phosphorylate p27 thereby targeting p27 for ubiquitylation and degradation^{43–47}. However, if T cells receive a TCR signal in the absence of CD28 co-stimulation, AKT and CDK2 are not activated, p27 is not degraded, and instead p27 is transcriptionally induced by TCR-coupled cyclic AMP signalling^{46,48}. Similarly, T cells that fail to progress through the cell cycle following TCR and CD28 co-stimulation do not downregulate p27, do not differentiate and are instead rendered anergic^{29,33}. Increased p27 levels are necessary and sufficient to prevent T cell differentiation and induce anergy, as forced expression of p27 in TCR and CD28 co-stimulated cells leads to transcriptional repression of the gene encoding IL-2 (REF. 46), and p27-deficient CD4⁺ T cells can differentiate into fully functional effector cells in the absence of CD28 co-stimulation both *in vitro* and *in vivo*^{47–49}. Together, these studies indicate that p27 is a sensor of co-stimulatory signals through CD28 that actively participates in establishing the anergic programme as an alternative cell fate from differentiation (FIG. 1).

CDK5 and CDK2 promote effector T cell differentiation. There are several potential molecular mechanisms to explain the T cell anergy-promoting activity of p27. It can inhibit the GTPase RHOA, which promotes actin depolymerization and motility in other cell types⁵⁰. However, the effect of p27 on cytoskeletal dynamics and motility in T cells has not been studied, and it is not clear how this function of p27 might explain the dominant loss of cytokine function exhibited by anergic T cells. p27 has also been shown to bind to the AP1 co-activator JAB1 (also known as SGN5 and CSN5), leading to inhibition of AP1-driven promoter activity, as shown using a reporter assay in transiently transfected cells⁴⁶. A loss of AP1 activity could contribute to defective IL-2 production, but a role for the inhibition of JAB1 by p27 in physiological T cell anergy has not been established. However, the fact that p27 is a CDK inhibitor implicates the CDK cascade in the control of anergy avoidance and T cell differentiation. This is supported by recent studies of CD4⁺ T cells deficient in active CDK2, the major target of p27 (REF. 51). Genetic, short hairpin RNA (shRNA)-mediated or pharmacological inhibition of CDK2 activity in CD4⁺ T cells during *in vitro* TCR and CD28 co-stimulation resulted in a twofold to threefold decrease in both IL-2 production and differentiation into IFN γ -producing effector cells⁵¹. Upon restimulation, CDK2-deficient T cells had a more marked defect in secondary IL-2 and IFN γ production, comparable to that of wild-type anergic T cells that were primed in the absence of co-stimulation. This suggests that in the absence of CDK2 T cells do not ‘realize’ that they have received co-stimulation and, as a result, undergo a default anergic response. This is the opposite phenotype to that of p27-deficient CD4⁺ T cells, which fail to sense a lack of co-stimulation and can differentiate in response to TCR signals alone. As described above, the interphase CDKs (such as CDK2) do not seem to have a requisite role in T cell division, which indicates that these effects are not simply a result of differences in cell division. These results position CDK2 downstream of the CD28 and IL-2 receptors as an anergy-avoidance sensor that promotes T cell differentiation (FIG. 1).

CDK5 has also been shown to participate in T cell activation and differentiation. CDK5 activity depends upon the atypical cyclin partner p35, which is highly expressed in neural tissue, such that mice deficient in CDK5 or p35 have severe defects in neuronal development^{52,53}. Neither of these

factors is expressed by resting T cells, but it was recently found that CDK5 and p35 are induced upon T cell activation⁵⁴. Conditional deletion of the genes encoding CDK5 or p35 in CD4⁺ T cells resulted in decreased activation in response either to agonistic TCR- and CD28-specific antibodies or to allogeneic dendritic cells⁵⁴.

How then do these CDKs regulate effector T cell differentiation? CDK5 does not regulate the cell cycle, but rather it is involved in actin polarization and chemokine-directed T cell migration *in vitro*, which indicates that CDK5 might regulate T cell activation and migration by phosphorylating factors involved in cell motility (BOX 3; FIG. 2). The function of CDK2 in promoting T cell cytokine production is also probably independent of its role in cell cycle regulation, as *Cdk2*-deficient mice exhibit normal T lymphopoiesis⁵⁵, and the dynamics of mature T cell cycle progression and division are normal in the absence of CDK2 (REF. 51). In addition, the effects of p27 or CDK2 on secondary T cell responsiveness are established by ~18 hours after primary stimulation⁵⁶ (A.D.W. and P.A.M., unpublished observations), a time point before the first G1 to S phase transition has occurred in activated

T cells²⁷. A more likely explanation is that CDK2 directly phosphorylates and regulates factors involved in TCR signalling and/or effector gene transcription. CDKs regulate a large number of transcription factors (such as SMAD3, JUN, SP1, RUNX1 and signal transducers and activators of transcription (STATs)) that in turn control cytokine gene expression (BOX 3; FIG. 2). Lineage-specific transcription factors (such as T-bet, GATA-binding protein 3 (GATA3) and retinoic acid receptor-related orphan receptor- γ t (ROR γ t)) also contain CDK phosphorylation motifs (S/T-P), but it is not known whether these factors are subject to regulation by the CDK pathway. Further studies will be required to definitively show that CDKs have a direct effect on T cell differentiation that is independent of any effects on cell division.

Regulating T cell memory

In addition to its role in the decision between T cell anergy and effector differentiation, p27 regulates the formation and persistence of memory during CD4⁺ and CD8⁺ T cell immune responses. A productive *in vivo* T cell response requires a clonal expansion phase during which most daughter cells undergo a terminal differentiation

programme, involving the elaboration of effector functions and apoptosis. However, some daughter cells adopt a memory precursor cell fate, characterized by decreased effector function and increased proliferative capacity (FIG. 1). These precursors survive the contraction phase of the response to establish the antigen-specific memory cell pool.

The CD4⁺ T cell response to viral infection is restricted by p27, as p27-deficient mice accumulate twofold to threefold more virus-specific CD4⁺ T cells than wild-type mice during the acute phase of LCMV infection⁵⁷. The effector phase of the LCMV-specific CD8⁺ T cell response is not quantitatively affected by a lack of p27 (REF. 58); however, both CD4⁺ and CD8⁺ T cells exhibit a delay in the contraction phase in p27-deficient mice, leading to a stable, long-term increase in the size of the memory T cell pool compared with wild-type mice. In the absence of p27, virus-specific T cells express lower levels of T-bet and KLRG1, which promote terminal effector differentiation, and higher levels of BCL-6, which favours the development of memory precursors^{57,59}. Upon secondary antigen encounter, p27-deficient CD4⁺ and CD8⁺ T cells have a more polyfunctional, central memory phenotype, producing high levels of IL-2 and undergoing a more marked burst of secondary clonal expansion than cells expressing p27 (FIG. 1). The KIP family member p21 (also known as CIP1; which is encoded by *CDKN1A*) also has a role in limiting the secondary clonal expansion of memory T cells⁶⁰. Both p21 and p27 inhibit CDK–cyclin complexes, but the extent to which they can compensate for each other in limiting memory T cell proliferation is not clear. It is possible that p21, which is highly expressed in activated memory T cells, has a non-redundant role owing to its unique ability to inhibit the DNA replication factor proliferating cell nuclear antigen (PCNA) during S phase.

How does p27 regulate memory formation? Part of the effect of p27 on CD8⁺ T cell memory depends on p27 expression in non-lymphoid tissue⁶¹. Dendritic cells derived from p27-deficient mice do not have enhanced T cell priming capacity⁶¹, and it is not clear how p27 activity in non-lymphoid tissues affects the number and quality of cytokine-producing memory CD8⁺ T cells. However, p27 clearly also has a cell-intrinsic role in controlling antigen-independent homeostatic cell division in both CD4⁺ and CD8⁺ memory T cells^{57,58}, which indicates that p27 and also p21

Box 2 | Alternative models to explain cell division as a force for T cell differentiation

Epigenetics

Epigenetic mechanisms are known to regulate T cell differentiation³⁵, but the extent to which these processes are coupled to cell division is not clear. The original idea that mitosis is required for epigenetic change came from an outdated view that histone and DNA methylation are enzymatically irreversible, and could only be changed passively through the synthesis of new unmodified daughter strands. However, it is now clear that the activity of methyltransferases is opposed by other enzymes (such as histone demethylases and TET hydroxylases⁹³), which questions the necessity of the cell cycle for epigenetic change. However, there is evidence that the cyclin-dependent kinase (CDK) cascade links the cell cycle with epigenetic processes, as CDK2 phosphorylates the linker histone H1, thereby inhibiting the binding of core nucleosomes and relaxing chromatin structure⁹⁴. This is thought to prepare chromosomes for decondensation before S phase, but could also regulate chromatin structure locally at effector T cell genes. CDK2 can be recruited to nuclear factor- κ B-responsive genes, where it regulates histone acetyltransferase activity. CDK2 also phosphorylates the H3K27 methyltransferase EZH2 (REF. 95) and the DNA methyltransferase DNMT1 (REF. 96), which promotes their gene silencing activity. In this way, CDK activity during cell cycle progression could directly lead to epigenetic change in T cells that promotes the expression of effector T cell genes.

Asymmetric cell division

Asymmetric division is a mechanism by which stem cells generate cellular diversity, and the fact that proteins important for lymphocyte function can segregate asymmetrically between daughter cells has intriguing implications for T cell fate determination. Although it has been shown that CD8⁺ T cell daughters that receive more T cell receptor complexes have greater effector function later in the response³⁸, the degree to which asymmetric cell division contributes more generally to lymphocyte differentiation is not clear. Interestingly, in budding yeast, every cell division is asymmetric, and phosphorylation of components of this machinery by the yeast CDK1 homologue is required for unequal protein segregation⁹⁷. Whether a similar role exists in higher eukaryotes is not known, but it is possible that the CDK cascade could directly regulate this process in T cells and could drive T cell differentiation through the asymmetric segregation of signalling molecules and transcription factors.

Box 3 | Substrates that couple the CDK cascade to effector gene expression

Cyclin-dependent kinase 5 (CDK5) positively regulates cytoskeletal reorganization and T cell receptor signalling through the phosphorylation of WAVE1, WAVE2, ephexin 1 and coronin 1A^{54,98}. CDK2 counteracts the immunosuppressive effects of transforming growth factor- β (TGF β) by phosphorylating and inhibiting the function of SMAD3, a major transducer of TGF β signals⁹⁹; CD4⁺ T cells expressing a mutant form of SMAD3 that lacks the CDK phosphorylation motifs are resistant to the induction of anergy by co-stimulatory blockade⁴⁹. JUN, which partners with FOS to form the transcription factor AP1, is a CDK target, and CDK-mediated phosphorylation of AP1 enhances its activity¹⁰⁰. CDK2 phosphorylates the transcription factor SP1 in late G1 phase, thereby increasing its DNA binding affinity¹⁰¹, and CDK2 is recruited to nuclear factor- κ B (NF- κ B)-responsive genes where it regulates the transcriptional co-activator p300 (REFS 102, 103). CDK-mediated phosphorylation of RUNX1, a positive regulator of interleukin-2 (IL2) transcription, blocks its interaction with co-repressor complexes¹⁰⁴. Signal transducer and activator of transcription (STAT) molecules are crucial drivers of lineage-specific gene expression in cytokine-driven T cell differentiation¹⁰⁵. The CDK8 component of the nuclear mediator complex was recently shown to phosphorylate STAT1, STAT3 and STAT5 within the transcriptional activation domain¹⁰⁶, resulting in proper nuclear localization, DNA binding and induction of interferon- γ (*Ifng*) gene expression. STAT3 is a known substrate of CDK5 in neurons, and it is likely that CDK5 and CDK2 can use the same proline-directed motif as CDK8 to promote STAT-driven cytokine gene expression by T cells. More work is required to understand how these cell cycle-regulated kinases influence T cell activation and differentiation, but current evidence indicates that the CDK cascade functions together with other signalling pathways that are generally associated with antigen receptor stimulation, including the mitogen-activated protein kinase (MAPK), phosphoinositide 3-kinase (PI3K), inhibitor of NF- κ B (IKK) and protein kinase C (PKC) cascades.

function to restrict the size of the memory T cell pool through their ability to inhibit CDK activity and cell cycle progression. Another mechanism by which p27 restricts memory T cell responses may involve the regulation of FOXO1, a Forkhead box transcription factor with a crucial role in the effector-to-memory cell transition in both CD4⁺ and CD8⁺ T cells^{62,63}. Strong pro-inflammatory signals activate the phosphoinositide 3-kinase (PI3K)–AKT pathway in effector T cells, resulting in the phosphorylation of FOXO1 on S256. This forms a binding site for 14-3-3 proteins, resulting in the sequestration of FOXO proteins in the cytoplasm and inhibition of its transcriptional function⁶⁴. Cessation of PI3K–AKT signalling following antigen clearance and resolution of inflammation enables FOXO1 to localize to the nucleus and participate in the establishment of memory gene expression patterns. CDK2 can phosphorylate FOXO1 on the neighbouring S249, which blocks 14-3-3 binding to the S256 phosphorylation site and promotes FOXO1 nuclear localization and function⁶⁵. An intriguing possibility is that dysregulated CDK2 activity in p27-deficient effector T cells results in a gain of FOXO1 function and the transition of supranormal numbers of effector T cells to memory T cells. In this way, p27 may, under normal circumstances, function to control the effector-to-memory T cell transition by restricting CDK2-mediated enhancement of FOXO1 nuclear activity. Such a role could also explain why p27

has a preferential effect on the frequency of IL-2-producing central memory T cells (compared with non-IL-2-producing effector memory T cells), which is the same subset that is most affected by FOXO1 deficiency⁶⁶.

Controlling T cell tolerance

As might be predicted from their role in the decision between anergy, differentiation and memory, CDKs and their inhibitors have a marked influence on the stability of self tolerance and susceptibility to the induction of organ transplant tolerance *in vivo*.

The CDK cascade in the regulation of self tolerance. The function of the CDK inhibitory protein p21 affects self tolerance in a complex manner. The absence of p21 in mice leads to decreased autoimmune pathology in the lupus-prone BXSB strain⁶⁷, whereas p21 deficiency results in increased autoimmune susceptibility on the C57BL/6 background^{68,69}. Early-onset autoimmunity in male mice of the BXSB strain is the result of a strong Y chromosome-encoded autoimmune accelerator locus (*Yaa*). *CDKN1A* (which encodes p21) is a well-known p53 target gene that can inhibit S phase progression through binding to both CDK1 or CDK2 and PCNA. In the context of DNA replication and damage, loss of p21 function results in inappropriate S phase exit before DNA repair, thereby engaging the intrinsic cell death programme²¹. Similarly, loss of p21 function in the

BXSB model leads to increased apoptosis of activated autoreactive lymphocytes, potentially explaining the observed reduction in autoimmune pathology in these mice⁶⁷. Therefore, the dominant role for p21 in BXSB mice, in which autoreactive lymphocyte proliferation is being driven by a distinct susceptibility allele, might be the inhibition of CDK and/or PCNA function.

Conversely, the absence of p21 on a non-autoimmune-prone background (such as C57BL/6 mice) leads to a distinct late-onset lupus-like disease in aged female animals, involving increased CD4⁺ T cell activation, antinuclear antibody production and glomerulonephritis^{68,69}. It is not clear from these studies whether immunopathology is accompanied by a gain of CDK function, but it is possible that p21 might affect other pathways in this setting. For example, a large pool of p21 in activated T cells is not associated with CDKs, but instead inhibits T cell activation by binding to and inhibiting the function of multiple mitogen-activated protein kinases (MAPKs)⁷⁰. However, a similar late-onset lupus-like disease has been reported in aged p27-deficient mice on a non-autoimmune-prone background⁷¹, which indicates that a loss of CDK control might be the common link between these two models.

If CDK inhibitors promote self tolerance, then CDKs should promote susceptibility to autoimmune disease. This is the case for the CDK5–p35 complex, which is involved in cytoskeletal remodelling at the immune synapse⁵⁴. In C57BL/6 mice, tolerance to the myelin sheath around the spinal cord can be broken by immunization with myelin-derived self antigens together with strong adjuvants, leading to autoimmune pathology of the central nervous system analogous to multiple sclerosis in humans. However, mice genetically deficient in *Cdk5* or *Cdk5r1* (which encodes the CDK5 cofactor p35) have decreased incidence and severity of autoimmune disease in this model⁵⁴. This phenotype was associated with decreased TCR-induced actin polarization, blunted T cell activation and a defect in migration across a chemokine gradient⁵⁴. CDK5 is therefore required to break self tolerance in this model, and it will be important to determine whether any interphase CDKs (CDK2, CDK4 or CDK6) are similarly involved in autoimmune disease susceptibility.

Control of peripheral tolerance by p27–CDK2. In addition to regulating tolerance to autoantigens, the CDK cascade also participates in the acquisition of peripheral

tolerance to alloantigens. Normal mice can be induced to accept fully MHC-mismatched organ transplants if their alloreactive T cells are deprived of both CD28 and CD40 co-stimulatory signals⁷². Similar to *in vitro*-anergized T cells, these tolerant T cells express high levels of the CDK inhibitor p27 (REF. 47). Induction of p27 is crucial to control alloreactive T cell proliferation and differentiation, as p27-deficient mice accumulate fivefold more IFN γ -producing T cells than wild-type transplant recipients, and are able to reject allogeneic transplants despite combined co-stimulatory blockade⁴⁷.

Blockade of either the CD28 or the CD40 co-stimulatory pathway alone can prolong allograft survival but, under these circumstances, T cell-mediated inflammation ultimately results in rejection⁷². Graft-infiltrating T cells express high levels of CDK2 and of CDK2-induced proteins, such as cyclin A2 (REF. 51), which indicates that the CDK pathway is highly active in these cells and may contribute to allograft rejection. Indeed, CDK2 is required for rejection in this model, as CDK2-deficient recipients accumulate fivefold fewer differentiated IFN γ -producing T cells and fail to reject their allografts⁵¹. Therefore, p27 and CDK2 have crucial and opposing roles in T cell differentiation versus tolerance in response to alloantigens, both *in vitro* and *in vivo* (FIG. 1).

CDK2 restricts regulatory T cell function.

It is clear that the CDK regulatory network influences conventional T cell anergy and effector function, but tolerance to both self antigens and alloantigens also requires the activity of regulatory T (T_{Reg}) cells⁷³. Less is known regarding the role of the CDK cascade in T_{Reg} cell biology, but recent findings indicate that CDKs and their inhibitors do indeed have a marked effect on the function of these cells. Genetic evidence suggesting a role for CDKs in negatively regulating T_{Reg} cell function was uncovered using an allograft tolerance model. Surviving grafts from CDK2-deficient recipients contained twice as many Forkhead box P3 (FOXP3)⁺ T_{Reg} cells as the rejecting grafts from wild-type recipients⁵¹, which indicates that CDK2 might have a role in controlling T_{Reg} cell-driven tolerance. Consistent with this, T_{Reg} cells lacking CDK2 had enhanced suppressive function, in terms of both inhibiting conventional CD4⁺ T cell proliferation *in vitro* and ameliorating T cell-mediated intestinal inflammation in a murine model of inflammatory bowel disease⁵¹.

How does CDK2 inhibit T_{Reg} cell function? CDK2 could contribute to the homeostasis of T_{Reg} cells, which proliferate extensively *in vivo*⁷⁴, but CDK2-deficient mice do not have an altered frequency of CD4⁺CD25⁺FOXP3⁺ cells in the thymus or the periphery, as compared with wild-type mice⁵¹. CDKs regulate several transcription factors that, in addition to those with general roles in T cells (BOX 3), are of particular relevance to T_{Reg} cells, such as SMAD3 and FOXO1 (BOX 4). However, a recent study found that the T_{Reg} cell lineage-defining transcription factor FOXP3 is itself regulated directly by CDK2 (REF. 75). Murine FOXP3 contains four CDK phosphorylation motifs (S/T-P sites) in the amino-terminal repressor domain. Two of these motifs (S19 and T114) are conserved in human FOXP3, which contains an additional three CDK motifs in the repressor domain and one at the carboxyl terminus. At least two of the murine motifs, S19 and T175, can be phosphorylated by CDK2 *in vitro*, and FOXP3 is phosphorylated on S19 in cells⁷⁵. Mutation of the four CDK motifs in mouse FOXP3 results in increased protein stability, suggesting that CDK2 normally functions to destabilize this factor⁷⁵. This mutant form of FOXP3 had an increased ability to induce CD25 and repress IL-2 expression when expressed in CD4⁺ T cells, which in turn were more able to suppress conventional T cell proliferation and T cell-mediated inflammatory disease than cells expressing wild-type FOXP3 (REF. 75). This FOXP3 mutation recapitulates the gain of function exhibited by CDK2-deficient T_{Reg} cells, which indicates that a dominant role for this kinase in T_{Reg} cells might be to regulate FOXP3.

Signals from receptors such as CD28 and IL-2R are strong inducers of the PI3K–AKT pathway in conventional T cells, which in turn drives the CDK cascade⁷⁶. Because CDK2 activity can destabilize FOXP3 at the protein level, and potentially at a transcriptional level by targeting FOXO1 (BOX 4), it follows that T_{Reg} cells must have mechanisms at their disposal to circumvent this pathway when their function is needed. T_{Reg} cells actively oppose PI3K–AKT activation through induction of the AKT phosphatase PHLPP1 (REF. 77), and a recent proteomic study showed that human FOXP3⁺ T_{Reg} cells have markedly reduced expression of CDK1, CDK2 and CDK6 compared with conventional CD4⁺ T cells⁷⁸. In addition, murine thymus-derived T_{Reg} cells have increased expression of p27, the inhibitor of CDK2, compared with conventional T cells (A.D.W. and P.A.M., unpublished observations), and the suppressive activity of human CD4⁺CD25⁺ T cells correlates with p27 expression levels^{79,80}. Transforming growth factor- β (TGF β) can induce FOXP3 expression in conventional CD4⁺ T cells⁸¹. This is also accompanied by strong induction of p27, which seems to be a crucial component of the differentiation programme for peripherally derived T_{Reg} cells, as p27-deficient conventional CD4⁺ T cells are defective in inducing and maintaining FOXP3 protein levels in response to TGF β and IL-2 (A.D.W. and P.A.M., unpublished observations). Thus, current data indicate that the T_{Reg} cell lineage targets the CDK cascade at multiple levels, and imply that inflammatory signals could operate through the activation of a PI3K–AKT–CDK2 pathway to inhibit T_{Reg} cell differentiation, stability and function (FIG. 3).

Box 4 | CDK substrates involved in T_{Reg} cell differentiation and function

By regulating transcription factors involved in Forkhead box P3 (FOXP3) gene expression, the cyclin-dependent kinase (CDK) cascade could influence regulatory T (T_{Reg}) cell differentiation and stability. Transforming growth factor- β (TGF β)-activated SMAD3 cooperates with the Forkhead transcription factors FOXO3A and FOXO1 to induce FOXP3 transcription in response to TGF β and interleukin-2 (IL-2) (REFS 107, 108). CDK2 and CDK4 can phosphorylate SMAD3 and negatively regulate its transcriptional activity⁹⁹, and therefore efficient FOXP3 induction would presumably require the inhibition of CDK activity. FOXP3 induction in conventional T cells is opposed by the T cell receptor (TCR)-coupled activation of AKT, which phosphorylates FOXO1 and sequesters it in the cytoplasm in a 14-3-3-dependent manner¹⁰⁹. Both CDK1 and CDK2 phosphorylate FOXO1 at a distinct site from AKT, and promote its function by blocking its interaction with 14-3-3 proteins⁶⁵. FOXP3⁺ T_{Reg} cells oppose the effect of AKT on FOXO1 by inducing the expression of PHLPP1, an AKT phosphatase that directly inhibits the activity of the phosphoinositide 3-kinase (PI3K)–AKT pathway⁷⁷. Another study found that phosphorylation of a CDK motif within the nuclear localization domain of FOXO1 can inhibit its nuclear localization and transcriptional activity¹¹⁰. An intriguing possibility is that CDK-mediated phosphorylation promotes the function of FOXO1 when it is also phosphorylated by AKT, but inhibits FOXO1 function when the PI3K–AKT pathway is not active. In this scenario, CDK activity would promote FOXO1 function in conventional T cells, but inhibit FOXO1 function in T_{Reg} cells. FOXO1 also induces p27 (REF. 111), forming an autoregulatory loop that could promote its own function and the expression and function of FOXP3 through inhibition of CDK2 activity (FIG. 3).

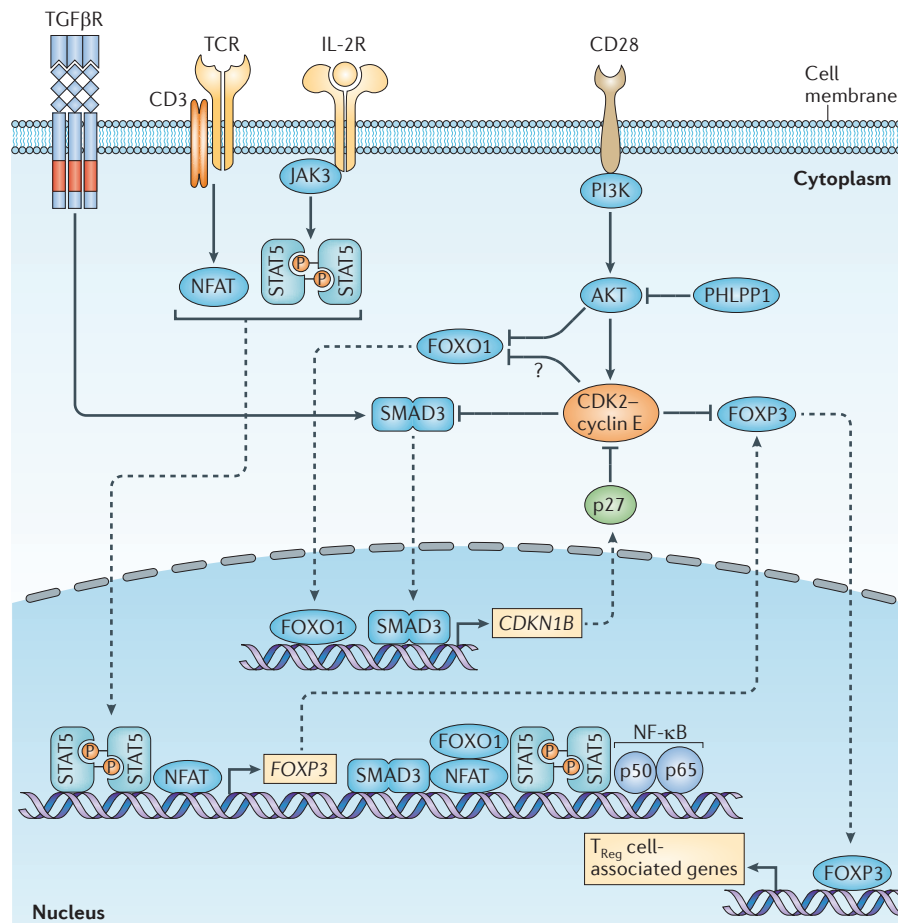


Figure 3 | Regulation of the regulatory T cell programme by CDK2 and p27. The receptors for antigen, transforming growth factor- β (TGF β) and interleukin-2 (IL-2) cooperate to induce expression of p27 (encoded by *CDKN1B*) and Forkhead box P3 (FOXO3) via transcription factors such as nuclear factor of activated T cells (NFAT), SMAD3, signal transducer and activator of transcription 5 (STAT5) and FOXO1. Concomitant activation of AKT and cyclin-dependent kinase 2 (CDK2), which inhibit the functions of SMAD3 and FOXO1, is opposed by PHLPP1 (an AKT phosphatase) and p27. This in turn protects FOXO3 from CDK2-mediated destabilization. JAK3, Janus kinase 3; NF- κ B, nuclear factor- κ B; PI3K, phosphoinositide 3-kinase; TCR, T cell receptor; T_{Reg} cell, regulatory T cell. Both positive and negative roles for CDK2 in FOXO1 function have been described (represented by a question mark). Dashed arrows indicate shuttling between the nucleus and the cytosol.

CDKs as immunotherapeutic targets

Our current model of the role of CDKs in driving specific transcriptional programmes involved in T cell differentiation and T_{Reg} cell function suggests that CDK inhibition could be a novel and potentially powerful strategy to treat T cell-mediated inflammatory diseases, such as autoimmunity and organ transplant rejection. In addition, as the interphase CDKs (CDK2, CDK4 and CDK6) are not absolutely required for cell cycle progression, specific targeting of these kinases should have fewer side effects than targeting the mitotic kinase CDK1 or other receptor-coupled kinase cascades.

There are several kinase-inhibitory drugs that have reasonable specificity for the CDK family and selectivity towards particular family members. Roscovitine (also known

as seliciclib and CYC-202 (Cyclacel)) targets CDK2 and is in clinical trials for the treatment of cancer^{82,83}. In experimental models, roscovitine has been shown to redirect T helper 17 (T_H17) cells to the peripherally derived T_{Reg} cell lineage by reinforcing FOXP3 expression⁸⁴, and also to ameliorate graft-versus-host disease in mice⁸⁵. In a model of systemic lupus erythematosus, aged NZB $\times$ NZW mice treated with roscovitine had less proteinuria, reduced IFN γ production by T cells and decreased lymphocyte proliferation⁸⁶. In rats, treatment with roscovitine decreased the proliferation of activated CD8⁺ T cells and the percentage of activated lymphocytes during a mixed-lymphocyte reaction, while inducing anergy in these cultures⁸⁷. In addition, treatment

with roscovitine prolonged the survival of fully MHC-mismatched kidney grafts compared with vehicle-treated controls⁸⁷. However, roscovitine also inhibits CDK7 and CDK9, which are important regulators of RNA Pol II-mediated transcription initiation and elongation, and it therefore causes complex and global changes in mRNA levels^{88,89}. At higher doses, roscovitine can inhibit the entire CDK family, and can have additional off-target effects on MAPKs. Purine bioisosteres of roscovitine can bind more specifically to CDK2 (REF. 90), and recent efforts have identified bioisosteres that are more potent than the parent drug, such as CVT313, H717 and purvalanols⁹¹.

There are currently 40 ongoing clinical trials of drugs that target various CDK family members in patients with cancer (24 Phase I trials, 15 Phase II trials and one Phase III trial), indicating that this is a feasible pathway to target clinically. Although they are not currently being evaluated against inflammatory disease in the clinic, CDK inhibitory drugs could provide an alternative to immunosuppressive modalities — such as mammalian target of rapamycin (mTOR) inhibitors, calcineurin inhibitors and tumour necrosis factor antagonists — which have serious complications including infection, neoplasia, hyperlipidemia and nephrotoxicity. Selective inhibition of CDK2 and/or CDK5 with current drugs or improved analogues could represent a novel treatment strategy for autoimmunity and organ transplant rejection.

Conclusions

The CDK cascade couples two processes that are absolutely required for the generation of an adaptive immune response: T cell clonal expansion and T cell differentiation. As such, CDKs and their regulators have crucial roles in T cell anergy, differentiation and memory, and in the development of autoimmunity and transplantation tolerance. The prevailing view positions CDKs as drivers of DNA replication and mitosis, providing a window of opportunity for the epigenetic remodeling of effector genes and/or the asymmetric segregation of effector molecules. However, a third potential model is outlined here. In this model, interphase CDKs become activated as T cells progress through the cell cycle and contribute directly to T cell differentiation by phosphorylating transcription factors involved in effector, memory and regulatory gene expression programmes (FIGS 2,3). The probability is that all three models are functional in proliferating T cells, and more work will be needed

to understand the complex ways in which the model described in this Opinion article might affect immune function. We propose that the CDK pathway is an antigen receptor-coupled cascade that functions alongside the MAPK, PI3K and inhibitor of nuclear factor- κ B (IKK) cascades to activate effector gene transcription and drive T cell fate decisions.

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doi:10.1038/nri3625

Published online 7 March 2014

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Acknowledgements

A.D.W. is a member of the Biesecker Pediatric Liver Disease Center at The Children's Hospital of Philadelphia, Pennsylvania, USA. Work relevant to this article by A.D.W. was supported by US National Institutes of Health grants AI054643 and AI070807, and work by P.A.M. was supported by a Goldie Simon Preceptorship Award sponsored by the Lupus Foundation of America Philadelphia Tri-State Chapter, Pennsylvania, USA, and the Pennsylvania Department of Health, USA.

Competing interests statement

The authors declare no competing interests.